# Lesson 1.2: Modular Arithmetic

**Learning Outcomes:** At the end of the lesson, the students are able to:

- rigorously define integers modulo $n$ as congruence classes;
- perform addition and multiplication modulo $n$;
- establish properties of modulo arithmetic.

## Congruence Classes

Let $m$ and $d$ be integers, where $d > 0$. If we enumerate the multiples of $d$ as $d, 2d, 3d, 4d, \ldots$ we know that this sequence of multiples is strictly increasing and there exists some $k \in \mathbb{Z}$ such that $kd > m$. Let $q$ be the largest integer such that $qd \leq m$. If we take $r := m - qd$, then we may write

$$m = qd + r,$$

where $r \in \{0, 1, \ldots, d - 1\}$. The following result is a generalization of this idea, which is quite handy for this course.

**Theorem 1.2.1 (Division Algorithm)**
Let $m, d \in \mathbb{Z}$, where $d \neq 0$. There exist unique $q, r \in \mathbb{Z}$ such that

$$m = qd + r,$$

where $0 \leq r < |d|$.

**Note** The integer $q$ is called the **quotient**, when $m$ is divided by $d$, while $r$ is called the **remainder**.

**Example 1.2.2** Let $m = 50$ and $d = 6$. Since $8 \cdot 6 = 36 < 50$ and $9 \cdot 6 = 54 > 50$, we take $q = 8$ and $r = 50 - 48 = 2$ so that

$$m = 50 = 8 \cdot 6 + 2 = qd + r.$$

Therefore, when 50 is divided by 6, the quotient is 8 and the remainder is 2.

If we instead let $m = -50$ and $d = 6$, we must take $q = -9$ and $r = 4$ so that

$$m = -50 = (-9) \cdot 6 + 4 = qd + r.$$

In this case, taking $q = -8$ does not work since if $m = qd + r$, this equation forces $r$ to be equal to $-2$, which contradicts the property that $0 \leq r < |d|$.

When dividing an integer by an integer $d$, it is a special ocassion when the remainder is 0. In this case, we may simply write $m = qd$, for some integer $q$ and say that $m$ is **divisible** by $d$ or $d$ **divides** $m$. We express this by writing $d \mid m$.

**Definition 1.2.3** Fixed an integer $n > 0$. We say that $a, b \in \mathbb{Z}$ are **congruent modulo** $n$ if they leave the same remainder when both are divided by $n$. In this case, we write $a \equiv b \pmod{n}$. The set of all integers congruent to $a$ modulo $n$ is called the **congruence class** of $a$ modulo $n$ denoted by $[a]$.

**Example 1.2.4** Let $n = 4$. Since 21 and 5 both leave a remainder of 1 when both are divided by 4, then $21 \equiv 5 \pmod{4}$. On the other hand, $12 \not\equiv 3 \pmod{4}$ since 12 leaves a remainder of 0, while 3 leaves a remainder of 3, when divided by 4. Figure 2 shows the set of all integers in the number line and two circles of the same color represent two integers which are congruent modulo 4.

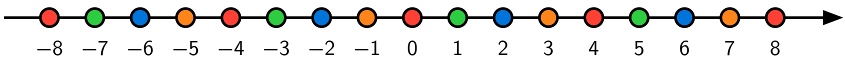


Figure 2: Congruence Modulo 4

As we can see, we may partition the set $\mathbb{Z}$ into four congruence classes namely $[0], [1], [2], [3]$. The red dots in Figure 2 represent the elements of $[0]$, the green dots represent the elements of $[1]$, the blue dots represent the elements of $[2]$, and the orange dots represent the elements of $[3]$.

The following result gives us a usefule criterion to determine whether two integers are congruent or not.

**Theorem 1.2.5** Let $a, b \in \mathbb{Z}$. Then $a \equiv b \pmod{n}$ if and only if $n \mid a - b$.

*Proof.* ($\Longrightarrow$) Assume that $a \equiv b \pmod{n}$. Then $a$ and $b$ leave the same remainder when both are divided by $n$. Thus, we may write

$$a = q_1 n + r \quad \text{and} \quad b = q_2 n + r,$$

for some integers $q_1, q_2$, and $r$, where $0 \leq r < n$. Hence,

$$a - b = (q_1 n + r) - (q_2 n + r) = (q_1 - q_2)n = kn,$$

where $k = q_1 - q_2 \in \mathbb{Z}$. This proves that $n \mid a - b$.

($\Longleftarrow$) Conversely, assume that $n \mid a - b$. Then there exists $k \in \mathbb{Z}$ such that $a - b = kn$. Thus, $a = b + kn$. By Theorem 1.2.1, we may write $b = qn + r$, for some $q, r \in \mathbb{Z}$, where $0 \leq r < n$. Hence,

$$a = b + kn = (qn + r) + kn = (q + k)n + r.$$

This implies that $a$ also has a remainder $r$, when divided by $n$. Therefore, $a \equiv b \pmod{n}$. This completes the proof. $\square$

**Example 1.2.6** Without checking individual remainders, we may now see immediately using Theorem 1.2.5 that

$$57 \equiv -15 \pmod{4},$$

since $57 - (-15) = 72$, which is divisible by 4.

Let $a \in \mathbb{Z}$. If we denote by $[a]$ the congruence class of $a$ modulo $n$, then by Theorem 1.2.5 we have

$$\begin{aligned} x \in [a] &\iff n \mid x - a \iff x - a = nk, \text{ for some } k \in \mathbb{Z} \\ &\iff n = a + nk. \end{aligned}$$

Thus,

$$[a] = \{a + nk \mid k \in \mathbb{Z}\}$$

consists of integers which are integer multiples of $n$ *away* from $a$.

**Remark 1.2.7** Congruence modulo $n$ is an *equivalence relation* on $\mathbb{Z}$. For all $a, b, c \in \mathbb{Z}$,

- (*reflexive*) $a \equiv a \pmod{n}$.

- (*symmetric*) $a \equiv b \pmod{n}$ implies $b \equiv a \pmod{n}$.
- (*transitive*) $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ imply $a \equiv b \pmod{n}$.

This equivalence relation allows us to partition $\mathbb{Z}$ into congruence classes mod $n$. Two integers are in the same class precisely when they are congruent modulo $n$. In general, there are exactly $n$ congruence classes of $\mathbb{Z}$ mod $n$ namely $[0], [1], [2], \ldots, [n-1]$, each corresponding to the remainder when an integer is divided by $n$.

**Definition 1.2.8 (Integers Modulo $n$)**
Let $n$ be a positive integer. The set of all **integers modulo** $n$ is defined as the set

$$\mathbb{Z}_n = \{[0], [1], [2], \ldots, [n-1]\}.$$

An integer modulo $n$ is **not** an integer. It is a representation of an entire class of integers which are congruent to one another modulo $n$. In many situations, the use of square brackets to represent an integer modulo $n$ can be cumbersome. We omit the square brackets for brevity and write

$$\mathbb{Z}_n = \{0, 1, \ldots, n-1\}.$$

when the context clear.

## Addition and Multiplication Modulo $n$

We now introduce binary operations on the set $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ which is based on addition and multiplication on $\mathbb{Z}$.

We define *addition modulo $n$* on $\mathbb{Z}_n$ as follows:

$$[a] + [b] = [a + b],$$

where $a + b$ is the sum of $a$ and $b$ as elements of $\mathbb{Z}$. We also define *multiplication modulo $n$* on $\mathbb{Z}_n$ as follows:

$$[a] \cdot [b] = [a \cdot b],$$

where $a \cdot b$ is the product of $a$ and $b$ as elements of $\mathbb{Z}$.

Before giving examples, we first prove that the definition above is independent of the manner we represent a congruence class.

**Theorem 1.2.9** In $\mathbb{Z}_n$,

1. if $[a] = [a']$ and $[b] = [b']$, then $[a + b] = [a' + b']$.
2. if $[a] = [a']$ and $[b] = [b']$, then $[a \cdot b] = [a' \cdot b']$.

*Proof.* We prove (1) and leave the proof of (2) to the student. Assume that $[a] = [a']$ and $[b] = [b']$. Then $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$. From Theorem 1.2.5, we have $n \mid a - a'$ and $n \mid b - b'$. Now, $(a + b) - (a' + b') = (a - a') + (b - b')$. Thus, $n \mid (a + b) - (a' + b')$. Hence, $a + b \equiv a' + b' \pmod{n}$. Therefore, $[a + b] = [a' + b']$. □

**Example 1.2.10** Let $n = 6$. By the definition of addition and multiplication modulo 5,

- $[2] + [4] = [2 + 4] = [6] = [0]$.
  The equality $[6] = [0]$ holds because $6 \equiv 0 \pmod{6}$.
- $[2] \cdot [4] = [2 \cdot 4] = [8] = [2]$.
  The equality $[8] = [2]$ holds because $8 \equiv 2 \pmod{6}$.

Using the brief notations introduced above, we can briefly write

$$2 + 4 = 0 \quad \text{and} \quad 2 \cdot 4 = 2.$$

Note that these equations hold contextually in $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ and should not be interpreted as ordinary operations on $\mathbb{Z}$. The complete Cayley tables for $\mathbb{Z}_6$ under addition modulo 5 and multiplication modulo 5 are presented in Table 5.

| + | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 2 | 3 | 4 | 5 |
| 1 | 1 | 2 | 3 | 4 | 5 | 0 |
| 2 | 2 | 3 | 4 | 5 | 0 | 1 |
| 3 | 3 | 4 | 5 | 0 | 1 | 2 |
| 4 | 4 | 5 | 0 | 1 | 2 | 3 |
| 5 | 5 | 0 | 1 | 2 | 3 | 4 |

| · | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 | 4 | 5 |
| 2 | 0 | 2 | 4 | 0 | 2 | 4 |
| 3 | 0 | 3 | 0 | 3 | 0 | 3 |
| 4 | 0 | 4 | 2 | 0 | 4 | 2 |
| 5 | 0 | 5 | 4 | 3 | 2 | 1 |

Table 5: Addition and Multiplication Modulo 6

With reference to Table 5, we see that

- $5 \cdot (4 + 3) = 5 \cdot 1 = 5$, and
- $(5 \cdot 4) + (5 \cdot 3) = 2 + 3 = 5$.

Thus, we have

$$5 \cdot (4 + 3) = (5 \cdot 4) + (5 \cdot 3).$$

As we may see, the distributive property of multiplication over addition holds for this particular example in $\mathbb{Z}_6$. As we shall soon see, this is not a mere coincidence.

The following remark allows us to compute in $\mathbb{Z}_n$ using the division algorithm.

**Remark 1.2.11** Note that in $\mathbb{Z}_n$, $a + b$ simply means the remainder when the integer $a + b$ is divided by $n$. Similarly, $a \cdot b$ equals the remainder when the integer $a \cdot b$ is divided by $n$.

The next theorem enumerates some of the algebraic properties addition modulo $n$ and multiplication modulo $n$ that holds among the elements of the set

$$\mathbb{Z}_n = \{0, 1, \ldots, n - 1\}.$$

**Theorem 1.2.12** Let $n > 1$ be a given integer. For all $a, b, c \in \mathbb{Z}_n$, we have

1. $a + b = b + a$ and $a \cdot b = b \cdot a$. *(commutative properties)*
2. $(a + b) + c = a + (b + c)$; and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$. *(associative properties)*
3. $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ *(distributive property)*

Notice that the properties listed in Theorem 1.2.12 holds true because addition and multiplication modulo $n$ are based on the standard addition and multiplication of integers, which inherently possess these algebraic properties. Furthermore, notice that

$$a + 0 = a \quad \text{and} \quad a \cdot 1 = a,$$

for all $a \in \mathbb{Z}_n$. Thus, we have the following result.

**Theorem 1.2.13** In $\mathbb{Z}_n$,

1. the element 0 is the identity element under addition modulo $n$;
2. the element 1 is the identity element under multiplication modulo $n$;
3. every element of $\mathbb{Z}_n$ has an *additive* inverse.

To see why statement (3) in Theorem 1.2.13 is true, let $a \in \mathbb{Z}_n$. If we define the element

$$-a := n - a$$

we see that

$$[a] + [-a] = [a] + [n - a] = [a + (n - a)] = [n] = [0].$$

It follows that $a + (-a) = 0$ in $\mathbb{Z}_n$.

**Example 1.2.14** In $\mathbb{Z}_6$, the additive identity is the element 0. From Table 5, we see that

- $0 + 0 = 0 \Rightarrow -0 = 0.$
- $1 + 5 = 0 \Rightarrow -1 = 5.$
- $2 + 4 = 0 \Rightarrow -2 = 4.$
- $3 + 3 = 0 \Rightarrow -3 = 3.$
- $4 + 2 = 0 \Rightarrow -4 = 2.$
- $5 + 1 = 0 \Rightarrow -5 = 1.$

For multiplicative inverses, it is more complicated than we can imagine. For example, we see that the multiplicative identity is the element 1. For reference, the Cayley table for multiplication modulo 6 is

- $0 \cdot a = 0$ for all $a \in \mathbb{Z}_6$, so 0 doesn't have a multiplicative inverse.
- $1 \cdot 1 = 1$, so the multiplicative inverse of 1 is 1.
- Looking at the row corresponding to the element 2, we see that $2 \cdot a \neq 1$, for all $a \in \mathbb{Z}_6$. So, the element 2 doesn't have a multiplicative inverse in $\mathbb{Z}_6$.
- Using a similar arguments, the elements 3 and 4 have no multiplicative inverse.

| $\cdot$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 | 4 | 5 |
| 2 | 0 | 2 | 4 | 0 | 2 | 4 |
| 3 | 0 | 3 | 0 | 3 | 0 | 3 |
| 4 | 0 | 4 | 2 | 0 | 4 | 2 |
| 5 | 0 | 5 | 4 | 3 | 2 | 1 |

- $5 \cdot 5 = 1$ implies that the multiplicative inverse of 5 is 5.

It follows that, only the elements 1 and 5 are invertible.

We shall address the invertibility of the elements in $\mathbb{Z}_n$ in a future lesson. For now, we simply note that under addtion modulo $n$, every element has an inverse. However, extra care must be observed when it comes to invertibility under multiplication modulo $n$.